

Urbanization Research — Case Study 01: Godagama 10200

Date: 2026-06-27

Case-study area: Godagama town, Colombo District, Western Province, Sri Lanka.

Repository: **Urbanization Research**. Godagama is the first city/town case study.

Public repo: <https://github.com/minzique/urbanization-research>

Purpose

This report models early urbanization pressure around Godagama using multiple evidence layers: Google-derived points of interest, OpenStreetMap history, official UDA/RDA planning documents, expressway/interchange context, developer market signals, and identified open datasets for future satellite/census expansion.

The report is data-led. Recommendations are included only where they follow directly from the measured pattern.

Executive findings

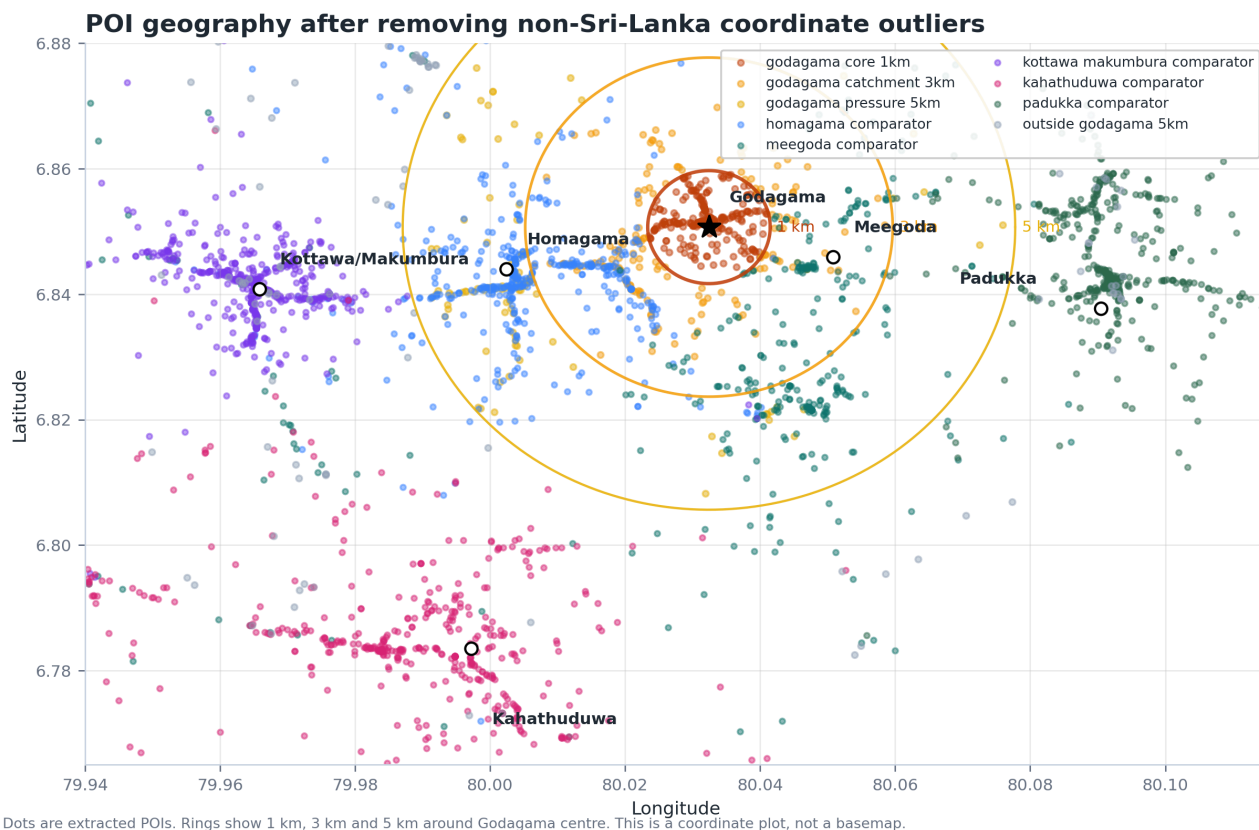
- Godagama has a dense core and a thinner surrounding catchment.** The 3 km catchment has **440 extracted POIs (15.6/km²)**. The 1 km core alone has **274 POIs**, about **87.2/km²**.
- The current Godagama POI layer is recent.** **82.4%** of dated Godagama POIs were first visible from **2021 onward**, the highest recent-share among the six comparator areas in this extract.
- Property/developer pressure is already visible.** Godagama has **63 property/developer POIs** in the 3 km catchment, **14.3%** of the extracted POI base. That share is close to Homagama and Kottawa/Makumbura.
- Comparator pressure is strongest in Kahathuduwa, then Meegoda and Kottawa/Makumbura.** Current score ranking: **Kahathuduwa (72.2)**, **Meegoda (57.6)**, **Kottawa/Makumbura (53.6)**, **Godagama (34.5)**.
- OSM history shows physical growth before full commercial maturity.** In the Godagama-area OSM bounding box, mapped buildings rose **+338%** from 2018 to 2026, while mapped road length rose **+33%**.

Evidence layers used

Layer	What it contributes	Status in this report	Main limitation
Google-derived POI extract	Current places, category mix, rough first-visible activity dates, chains/anchors	Quantitative: 3,494 geospatially valid POIs after removing one non-Sri-Lanka coordinate outlier	Earliest photo/activity date is not a verified opening date; review count and first review date not decoded yet

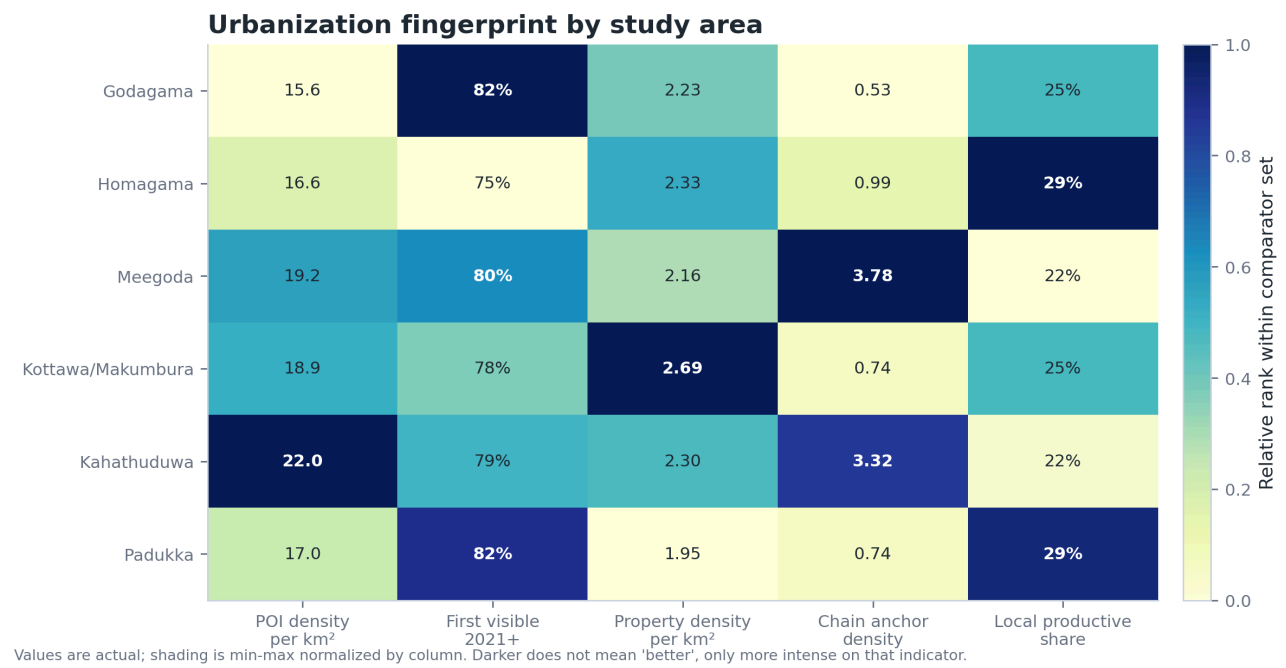
Layer	What it contributes	Status in this report	Main limitation
OpenStreetMap / ohsome history	Mapped buildings, roads, shops, food/drink places over time	Quantitative baseline: 2014–2026 time slices, charted as indexed growth	Affected by mapping completeness and edit bursts
UDA Homagama Development Plan	Official zoning, town hierarchy, environmental control zones, guide-plan areas	Integrated as planning evidence: Godagama HD Commercial Zone III; Kahathuduwa HD Commercial Zone II; wetland/paddy controls	Needs GIS polygon extraction for parcel-level analysis
RDA / Exway / gazettes	Expressway chronology, active interchanges, official corridor status	Integrated as transport context: OCH, Southern Expressway, Central Expressway comparator nodes	Needs exact GIS alignment/interchange layers in the model
Developer market signals	How land developers market highway access and future value	Integrated qualitatively and via Google land-development POIs	Need full land-price/time-series scrape for price model
DCS Census Portal / district handbooks	Population, housing, labour, local-authority context	Source identified; not yet numerically integrated	Next pass should export DS/GN-level 2012 vs 2024 fields
GHSL / WorldPop / VIIRS	Satellite built-up, settlement growth, night-time economic intensity	Source stack identified for next model expansion	Not yet extracted into the Godagama case-study calculations

Study geography



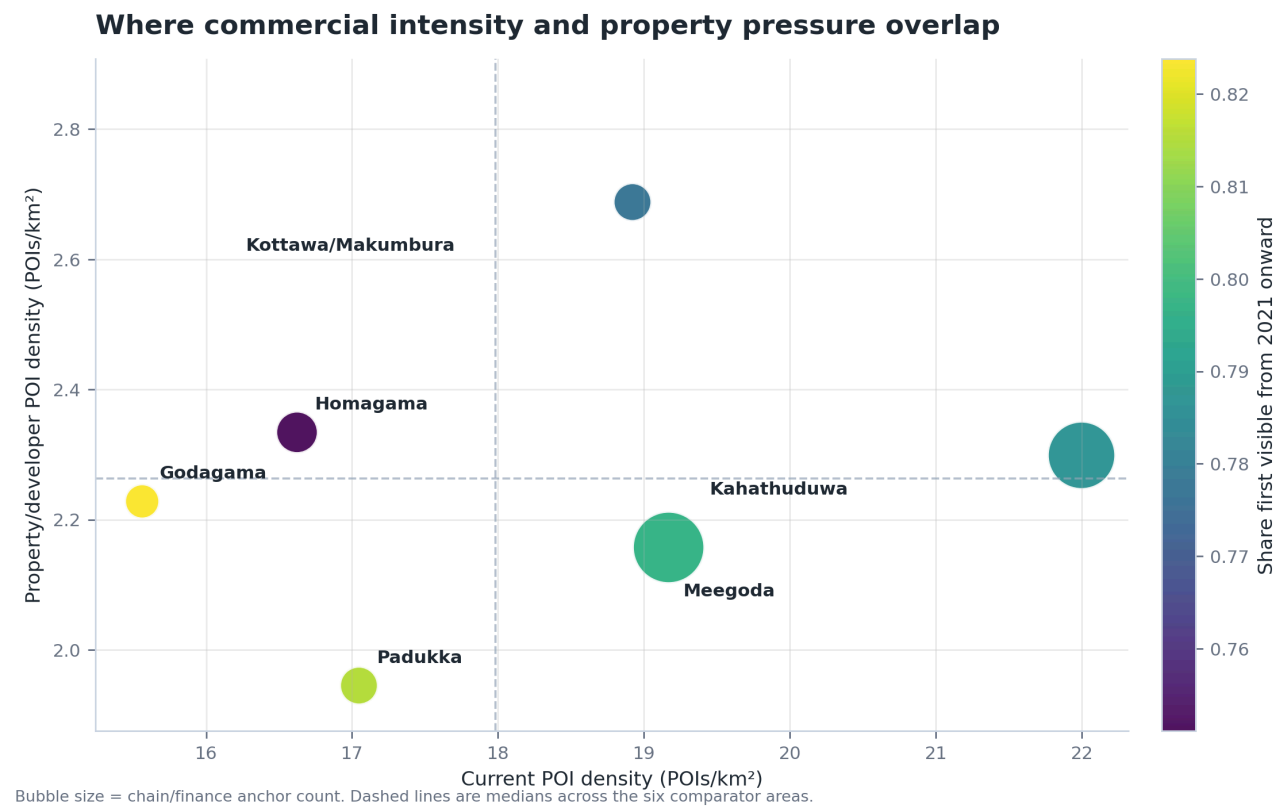
Urbanization fingerprint

Darker cells show which places are high on each measured indicator; printed values are actual values.



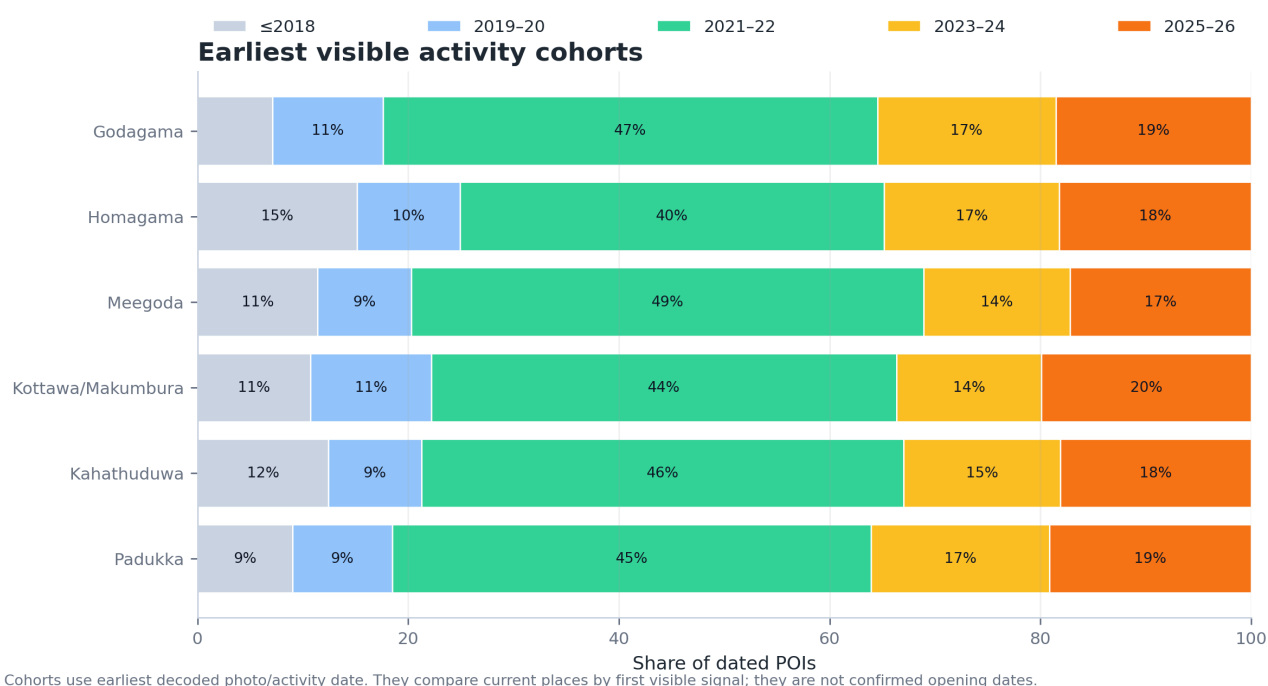
Pressure model

The score is a triage index, not a land-price forecast. Inputs: 35% POI density, 25% 2021+ first-visible share, 25% property/developer density, and 15% chain/finance anchor density.



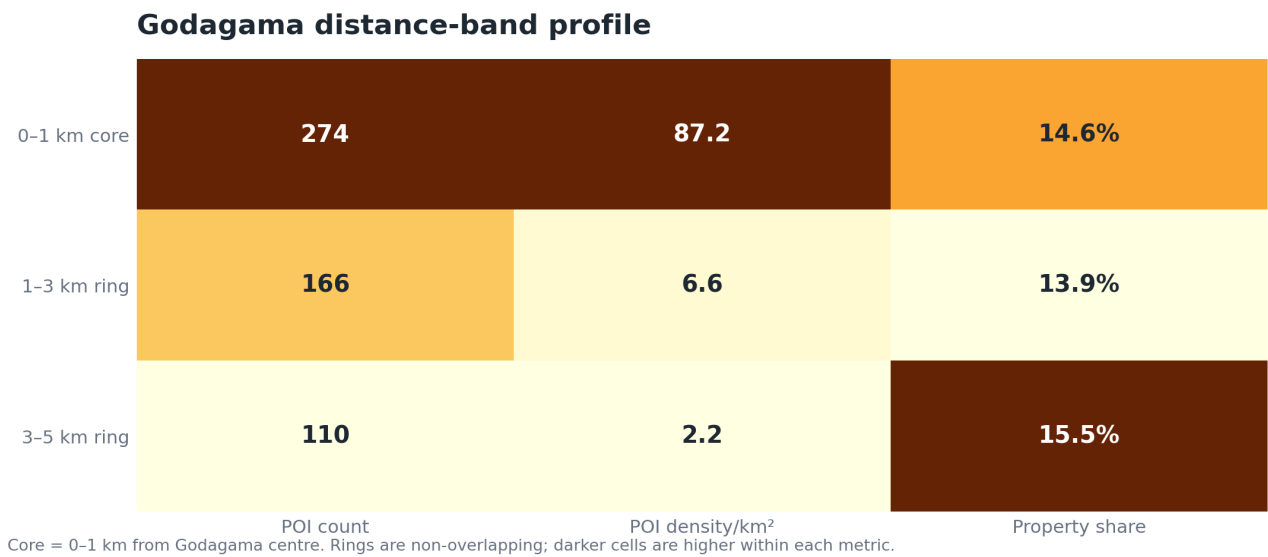
Area	POIs	POIs/km ²	Visible 2021+	Property share	Property POIs	Productive-service POIs	Score
Godagama	440	15.6	82.4%	14.3%	63	111	34.5
Homagama	470	16.6	75.1%	14.0%	66	137	21.0
Meegoda	542	19.2	79.7%	11.3%	61	117	57.6
Kottawa/Makumbura	535	18.9	77.8%	14.2%	76	135	53.6
Kahathuduwa	622	22.0	78.7%	10.5%	65	136	72.2
Padukka	482	17.0	81.5%	11.4%	55	138	31.2

Timeline signal



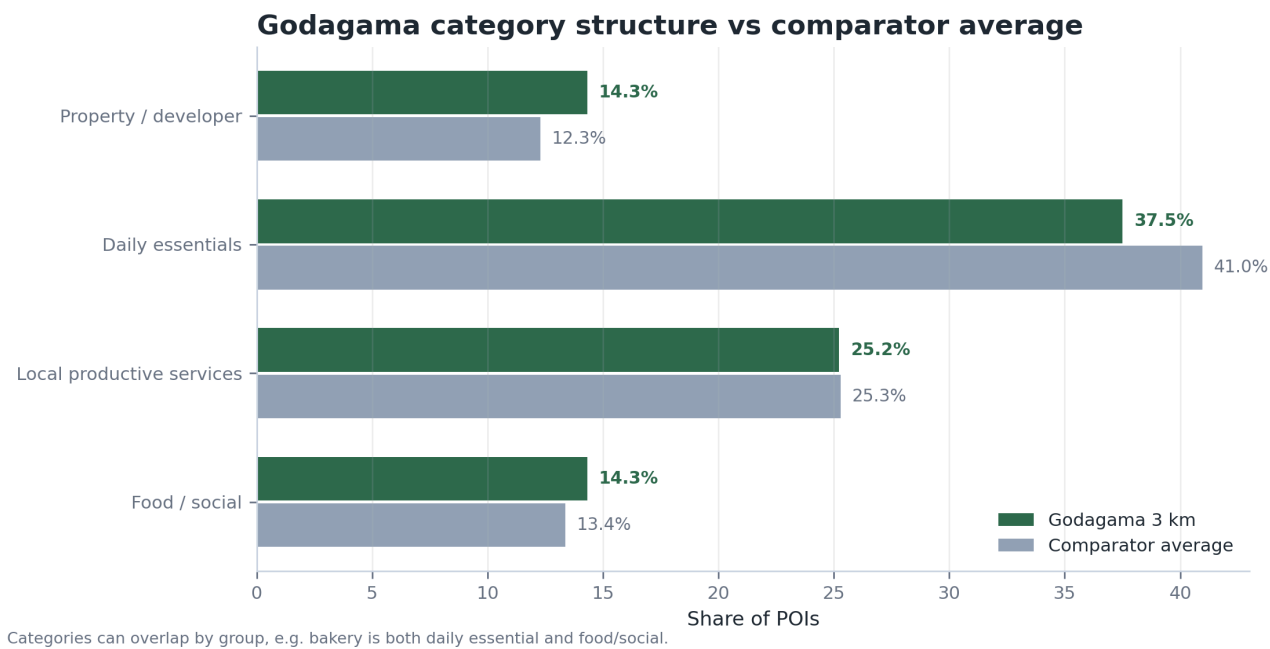
The 2021–2022 band is large across all areas. Treat it as a combined real activity + Google photo coverage signal. The comparison remains useful because the same extraction method is applied to every area.

Godagama distance-band structure



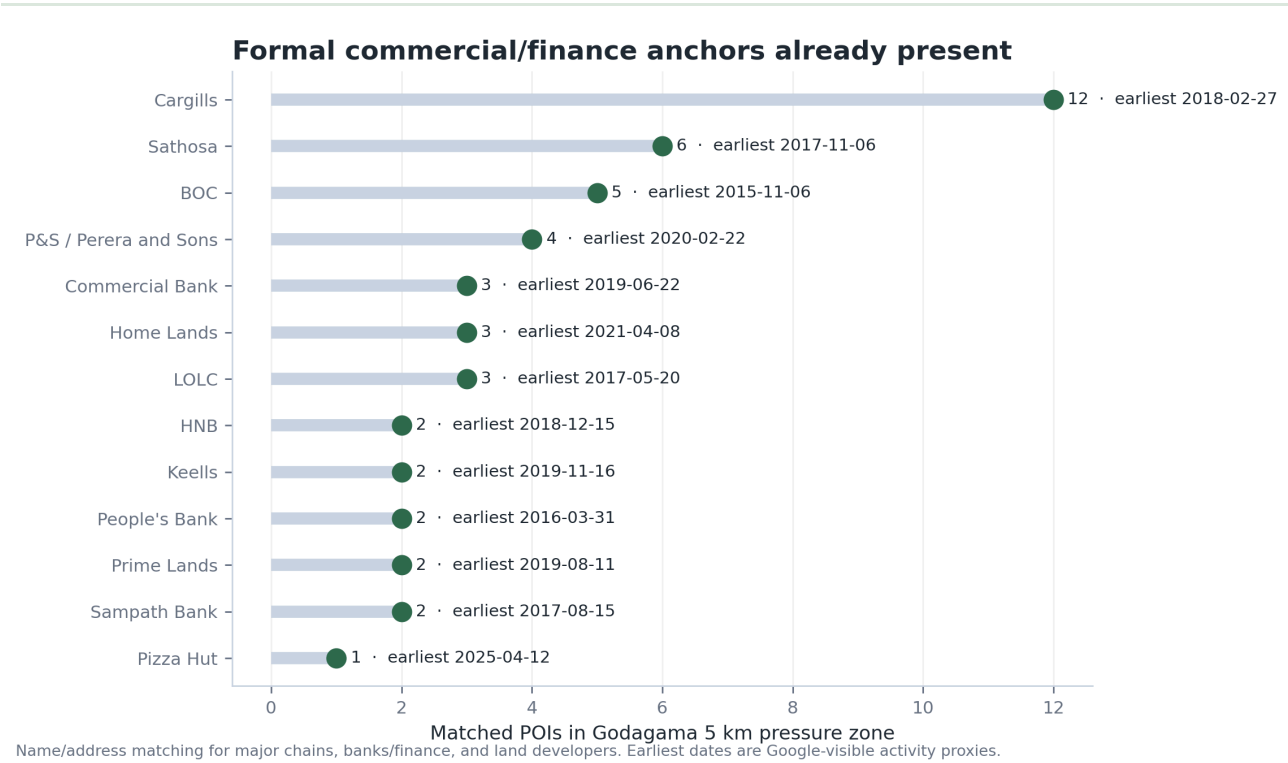
The 1 km core carries most of the visible density. The 1-5 km rings have lower density but still show property/developer pressure, so the town needs both core management and outer-ring growth guidance.

Category structure



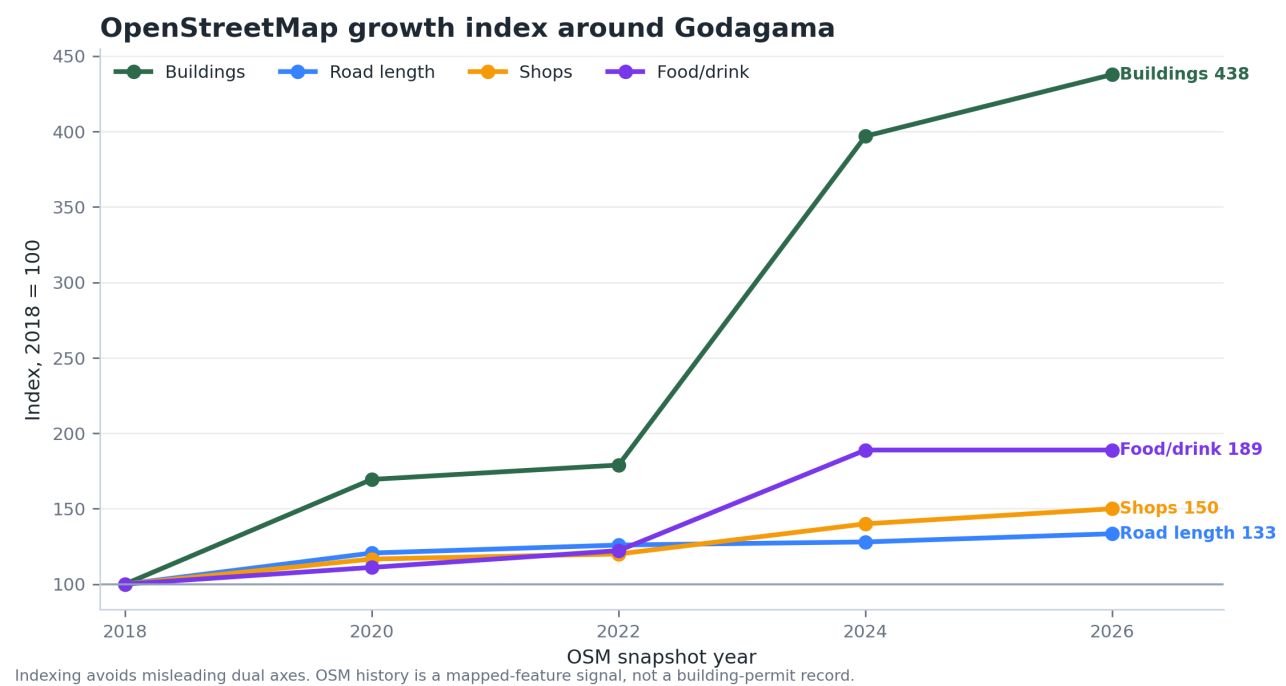
Godagama is not only a food or commuting suburb. The extracted mix shows daily essentials, clinics, hardware, vehicle service, electronics, courier, apartments, and land development. That supports a practical local-enterprise framing rather than a single-sector framing.

Chain, finance, and developer anchors



Anchor	Matched POIs in Godagama 5 km	Earliest visible date
Cargills	12	2018-02-27
Sathosa	6	2017-11-06
BOC	5	2015-11-06
P&S / Perera and Sons	4	2020-02-22
Commercial Bank	3	2019-06-22
Home Lands	3	2021-04-08
LOLC	3	2017-05-20
HNB	2	2018-12-15
Keells	2	2019-11-16
People’s Bank	2	2016-03-31
Prime Lands	2	2019-08-11
Sampath Bank	2	2017-08-15
Pizza Hut	1	2025-04-12

Physical growth baseline from OSM



OpenStreetMap history is not a building-permit record, but it shows the direction and scale of mapped physical growth. The 2018–2026 building and road signals are consistent with subdivision/build-out preceding a mature commercial layer.

Official planning and expressway context

Finding	Evidence collected	Planning/model implication
Godagama is not just passive sprawl	UDA Homagama plan identifies Godagama as High-Density Commercial Zone III	Treat Godagama as an official intensification node
Kahathuduwa is a stronger interchange comparison	UDA plan identifies Kahathuduwa as High-Density Commercial Zone II and a guide-plan area; Kahathuduwa interchange is within Homagama PS	Use Kahathuduwa as the near-future comparator for interchange-driven growth
Highway access shapes market language	Prime Lands markets Dagny Godagama through Makumbura/Athurugiriya access; Landify Homagama through Kottawa hub/highway exit; Home Lands markets projects through interchange proximity	Developer listings are early-warning indicators for land-price and subdivision pressure
Environmental constraints are explicit	UDA plan includes Wetland Nature Conservation Zone and Paddy/Wetland Agricultural Zone; sensitive commercial projects trigger environmental/planning clearance	Map protect / guide / intensify zones before approving outer-ring growth
Central Expressway is a comparator, not the main Godagama driver	CE Phase II nodes such as Mirigama/Kurunegala are useful examples of interchange-led peri-urban land products	Use CE nodes to compare how land products appear before full commercial maturity

Developer market signals collected

Developer / project signal	Extracted evidence	Use
Prime Lands — Dagny Godagama	Marketed as rapidly developing, near High Level Road, Godagama Junction, Meegoda Economic Centre, Makumbura/Athurugiriya access; advertised at LKR 1,240,000 per perch upward	Direct Godagama land-market pressure signal
Prime Lands — Landify Homagama	Marketed near Homagama town, Kottawa Multimodal Hub, Kottawa Highway Exit; advertised at LKR 2,250,000 per perch upward	Comparator price/access signal
Prime Lands — Ever Green Kahathuduwa	Marketed through Kahathuduwa interchange access	Interchange-led comparator signal
Home Lands inventory/API	Projects collected for Meepe, Kahathuduwa, Athurugiriya/Panagoda, Mirigama with block counts, availability, prices/access claims	Useful for absorption and price-pressure model

Data-driven planning suggestions

1. **Core 0–1 km:** manage as a town centre. Priorities: crossings, drainage, frontage rules, parking/loading, shade, and small-shop continuity.
2. **Ring 1–3 km:** reserve room for local productive services: hardware, repair, vehicle service, electronics, courier, clinics, food, bakeries, and plant/garden businesses.
3. **Ring 3–5 km:** require drainage, paddy/wetland screening, road-connectivity review, and developer contribution before large approvals.
4. **Quarterly monitoring:** rerun the POI extraction and compare Godagama against Kahathuduwa and Kottawa/Makumbura.
5. **Next data pass:** add review counts/first-review dates, official opening dates for top anchors, land-price-per-perch data, UDA zoning polygons, GN/DS census fields, and GHSL/WorldPop/VIIRS raster metrics.

Public repository access

Public repo: <https://github.com/minzique/urbanization-research>

Display title: **Urbanization Research**

First case folder: `cases/godagama-10200/`



Scan for the public repository, source notes, report assets, and case-study data.

Public contents include:

- printable PDF report;
- source markdown report;
- cleaned charts;
- derived summary CSV/JSON and the extracted public POI dataset;
- source notes, methodology, and reproducibility scripts;
- no private session files, raw browser captures, WhatsApp data, or unrelated private project materials.